

Fidelis Deception®: Unified Active Deception Across On-Prem, Endpoint, Network, Cloud, and Hybrid Environments

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Key Takeaways

- Traditional detection tools generate high alert noise, while Fidelis Deception® creates zero-false-positive signals by triggering only when attackers interact with decoys or fake credentials.
- Credential-based attacks dominate breaches with 22% as entry points and ~292-day detection timelines, making early in-network detection critical.
- Deception technology places realistic decoys and breadcrumbs across endpoints, Active Directory, network, and cloud to lure attackers during reconnaissance and lateral movement.
- Unified coverage across on-prem, cloud, OT, and endpoints eliminates blind spots where attackers typically pivot between environments.
- Integrated with XDR, Fidelis Deception® enables instant detection, full TTP visibility, and automated containment before attackers reach real assets.

Deception technology that spans every layer of your enterprise and turns attacker presence into an instant, confirmed signal.

Here is something most security teams know but rarely talk about openly. Your tools already detect plenty. The problem is they detect everything, including thousands of things

that turn out to be nothing. By the time a real threat surfaces, it is buried in noise.

That is not a staffing problem. It is a signal quality problem. And it is exactly why cyber deception technology has moved from niche experiment to mainstream cyber defense strategy in 2026.

[Fidelis Deception](#)[®] takes a different approach. Instead of generating more alerts for analysts to review, it plants deception decoys, decoy systems, and false credentials throughout your environment, across on-premises, endpoints, network, cloud, and OT, and waits. No legitimate user ever touches a decoy. When something does, that is a confirmed attacker. Every time.

Metric	Value
Avg US data breach cost (2025)	\$10.22M
Days to detect a credential breach	292 days
Breaches using stolen credentials	22%
False-positive alerts per SOC per week	9,854

Why Your Current Tools Keep Missing the Attacker Already Inside

This is the scenario that keeps CISOs up at night. An attacker gets in, usually through stolen credentials, and spends weeks or months moving quietly through the network. They use legitimate accounts. They use built-in OS tools. Nothing looks wrong.

The Verizon 2025 Data Breach Investigations Report², which analyzed over 22,000 incidents, found that 22% of all breaches started with stolen credentials, still the single most common attack vector. In 88% of web application attacks, stolen credentials were the primary method used to gain unauthorized access.

Now add the detection timeline. IBM's 2024 Cost of a Data Breach Report¹ found that credential-based breaches take an average of 292 days to identify and contain. That is nine and a half months. The attacker has the run of your environment for that entire window.

Why so long? Because the tools watching your environment were not built for this scenario. They flag anomalies, but an attacker using a valid account and standard Windows tools does not look anomalous. It looks like IT.

Then there is the alert problem. According to Ponemon Institute research⁴, the average SOC team receives 22,111 alerts every week. Roughly 9,854 of those are false positives.

Analysts spend around 25% of their working hours chasing them. When the real threat arrives, it lands in the same queue.

[Cyber deception](#) technology does not try to improve this noise problem incrementally. It solves it structurally. If no legitimate user ever touches a fake asset, then any interaction with one is 100% confirmed malicious. There is nothing to triage.

What Deception Technology Actually Does and Why It Works

The core principle is simple: seed your environment with fake assets that look real and wait for an attacker to find them. The execution is what separates effective deception platforms from basic honeypots.

Modern cyber [deception technology](#), Fidelis Deception[®], starts by profiling your actual environment. Operating systems, open ports, running services, Active Directory structure, network topology. Only then does it deploy decoys, because decoys that do not match your real terrain get spotted and ignored by skilled attackers.

Two layers work together:

Decoy Systems and Fake Assets

Decoy systems are full-fidelity fake servers, databases, endpoints, and network devices. A decoy mimics legitimate servers right down to the OS, open ports, and running services an attacker would see on a network scan. When an attacker probing for vulnerable systems connects to a decoy, every action is captured, including commands, credentials attempted, protocols used, and lateral paths tried.

Breadcrumbs and Fake Credentials in Active Directory

Breadcrumbs are placed on real endpoints. Fake credentials sit in browser password stores, registry keys, and cached network shares, all pointing toward decoy systems. When an attacker compromises a workstation and starts mapping pathways to higher-value systems, the breadcrumbs guide them straight into a trap.

[Inside Active Directory, Fidelis Deception[®]](#) seeds fake accounts, fake groups, and honeytokens. An attacker running Kerberoasting or pass-the-hash will inevitably touch them. The moment they do, security teams know the attacker's location, account used, and intended next hop, all before any real asset is reached. This early warning gives defenders the advantage.

Advanced Deception Technology Comparison

- Real-World Performance Data
- Avoiding False Savings

- Why Fidelis Outperforms the Competition

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How Fidelis Deception[®] Catches a Credential-Based Attack: Step by Step

This is the attack scenario that defeats most detection tools. Here is how deception technology changes the outcome:

Step 01: Attacker gains initial access via stolen credentials

Valid credentials look like a valid user. No perimeter alert fires. The attacker authenticates silently and begins mapping the environment.

Step 02: Reconnaissance and Active Directory enumeration begin

The attacker queries AD for high-value accounts, scans the network, and checks shared drives. Fidelis breadcrumbs and fake AD entries are already seeded here.

Step 03: Attacker follows a breadcrumb toward a decoy

A planted fake credential or network share points toward a decoy server. The attacker moves toward it, believing it is a real high-value target.

Step 04: Fidelis fires a zero-false-positive alert with full TTP context

Every command, credential attempt, and protocol used is logged. Security analysts receive one confirmed alert, not 500 alerts to triage.

Step 05: Fidelis Elevate[®] XDR auto-isolates the threat

The compromised session and affected segment are quarantined automatically. Lateral movement stops before any real asset is reached.

Coverage Across Every Layer: On-Prem, Endpoint, Network, Cloud, OT

Gaps in deception coverage become the routes attackers learn to use. An intruder who pivots from a cloud workload to an on-premises server, or from corporate IT to an OT segment, evades decoys that cover only one layer.

Fidelis Deception[®] covers every layer from a single centralized deception server. All telemetry flows into one management console. Here is what that looks like across each environment:

Environment	What Fidelis Deploys	Threats Detected
On-Premises	Decoy servers, databases, file shares built from your actual terrain	Lateral movement, insider threats, privilege escalation
Endpoint	Fake credentials, planted browser passwords, registry breadcrumbs	Account hijacking, credential harvesting, pass-the-hash
Network	Decoy services across DNS, TCP, HTTP, SSL, and custom app protocols	Unauthorized reconnaissance, port scanning, lateral movement. Provides early threat detection across the network fabric.
Cloud (AWS)	Fake IAM entries, decoy storage buckets, cloud-native trap resources	Cloud credential abuse, cloud-native lateral movement
OT / ICS	Decoy ICS devices running industrial protocols (Modbus, DNP3, etc.)	Recon against industrial control systems and SCADA
Active Directory	Fake accounts, fake service principals, honeytokens inside AD	AD enumeration, Kerberoasting, credential theft at recon stage

On-Premises and Network Deception

On-prem decoys are auto-generated by profiling your actual servers, workstations, and services. They look authentic because they are built from your real cyber terrain. [Network deception](#) extends this across every protocol, including DNS, TCP, HTTP, and SSL, so any

unauthorized reconnaissance or lateral movement that touches the network fabric is captured with early threat detection applied to real assets.

Endpoint Deception

Endpoints are where lateral movement begins. Planted fake credentials and breadcrumbs on real machines guide attackers who have compromised a user account away from real assets and toward decoys. [Endpoint deception](#) is especially effective against account hijacking attacks. It catches the attacker at the movement stage, before they find anything of value.

Cloud Deception

IBM's 2024 Cost of a Data Breach Report found that 40% of breaches involved data across multiple environments, with those breaches costing more than \$5 million on average and taking 283 days to contain.

Fidelis Deception[®] [extends into AWS with cloud-native trap resources](#), including fake IAM roles, decoy storage buckets, and phantom cloud databases, built to attract attackers who have compromised a cloud workload and are pivoting deeper into the environment.

Deception for OT and Industrial Control System Environments

OT and ICS environments run legacy systems that cannot support traditional security agents. Yet these environments are increasingly exposed. CISA's Industrial Control Systems advisory data shows a persistent rise in cyber incidents targeting critical manufacturing, energy, and utilities since 2023.

[Deception for OT environments](#) requires a different approach. Fidelis deploys decoy ICS devices using the actual industrial protocols those environments use, including Modbus, DNP3, and EtherNet/IP. Legitimate ICS components do not send traffic to unauthorized systems. Any interaction with an ICS decoy is, without exception, an attacker probing your operational technology.

Fidelis Deception[®] vs. Legacy Honeypots: A Direct Comparison

Security teams sometimes ask whether modern advanced deception technology is just honeypots with better marketing. It is not. The differences are structural.

Capability	Legacy Honeypots	Fidelis Deception [®]
Coverage	Perimeter only	On-prem, endpoint, network, cloud, OT, AD

Capability	Legacy Honeypots	Fidelis Deception [®]
Authenticity	Generic and often detectable	Auto-generated from your real cyber terrain
Alert quality	High noise, many false positives	Zero false positives. Every alert is confirmed.
Active Directory	Not supported	Native fake AD entries and honeytokens
XDR integration	None	Native integration with Fidelis Elevate[®] XDR
Auto-response	Manual investigation only	Automatic isolation and containment
TTP capture	Basic logs	Full attacker path: commands, credentials, tools
OT / ICS support	Not supported	Industrial protocol decoys supported

The core gap: a traditional honeypot at the perimeter catches the occasional unsophisticated external probe. Advanced deception technology catches the attacker who already bypassed the perimeter and is moving through your network right now, protecting real assets before they are ever reached. That is the threat that causes the damage.

How Deception Technology Solves Alert Fatigue for Security Teams

Alert fatigue is not a volume problem you solve by adding headcount. It is a signal quality problem. When analysts cannot trust their alerts, they begin filtering them mentally, and that is when real threats get through.

Deception inverts this. Every decoy interaction is confirmed malicious. No legitimate process touches a fake credential. No legitimate user connects to a phantom database. When Fidelis fires an alert, it means one thing: an attacker is in the environment right now.

The practical result for security teams:

- Fewer total alerts, but all of them demand immediate attention
- No false positives from decoy interactions, ever
- Full tactics, techniques, and procedures (TTP) context attached to every alert before the analyst opens it

- Faster mean time to detection. The 292-day credential breach window collapses to hours.
- [Automated response](#) through Fidelis Elevate® XDR stops lateral movement without a manual triage step

"With Fidelis Deception®, we're changing the rules of the game. Now we have the attackers running for cover because they understand that we can find them even if they managed to bypass our perimeter."

Head of IT Security, Fortune 1000 Pharmaceutical Company

Detecting Compromised Users, Credential Theft, and Insider Threats

These three threat types share a common problem: the attacker looks legitimate. A compromised user account interacts with systems it has permission to access. An insider threat operates through normal access channels. [Credential theft](#) hands the attacker a valid identity.

Conventional tools struggle here. Deception techniques do not, because they set a trap that legitimate users never walk into.

Credential Theft and Account Hijacking

IBM's 2025 Cost of a Data Breach Report puts the average US breach cost at \$10.22 million, a record figure. Credential-based breaches account for a significant portion, carrying that nine-month detection timeline.

Fidelis plants fake credentials inside Active Directory and on endpoints. The moment an attacker uses one, to authenticate to a decoy server, open a phantom file share, or attempt privilege escalation, the platform fires instantly. No waiting nine months.

Detecting Compromised Users Through Lateral Movement

As an attacker moves through the environment, they scan, probe, and test every system they can reach. Fidelis breadcrumbs are embedded at every stage of this movement. Each interaction is logged. Security teams receive a precise map of the attacker's path: where they came from, what they touched, what credentials they tested, and where they were heading.

This is adversary behavior observed in your specific environment, targeting both legitimate assets and decoys. That specificity is what makes the [intelligence actionable against advanced persistent threats](#).

Catching Insider Threats

An [insider threat](#), or a compromised account with legitimate access, is nearly invisible to tools that rely on behavior baselines. Fidelis places deception decoys in areas where legitimate assets exist but where users have no reason to go. If an account touches one of those systems outside its normal operating boundary, that is an immediate signal: the account is compromised or the user is acting maliciously.

Stopping a Successful Privilege Escalation Attack

[Privilege escalation](#) is often the final step before an attacker reaches and damages business-critical systems. Fidelis plants decoy administrative accounts and fake high-privilege credentials that appear exactly like the kind of access an escalating attacker is looking for. Any attempt to use them triggers a confirmed alert and an automated response.

Deception Technology in a Zero Trust Environment

Zero trust limits what attackers can do once they are inside your network. Deception confirms when they are inside at all. The two principles are designed to work together, not compete.

In a zero trust environment, legitimate users follow expected access paths to expected systems. Deception decoys sit outside those paths entirely. No legitimate user encounters them. An attacker with valid stolen credentials almost always does, because they are exploring terrain they have no map for.

The combination converts zero trust from a passive architectural posture to an active defense strategy. Zero trust limits damage. Cyber deception technology detects that damage is being attempted.

How Fidelis Deception® Provides Valuable Intelligence on Attacker Behavior

Detection is not the only output. Every decoy interaction is an intelligence event.

When an attacker engages with a Fidelis decoy, the platform captures their complete tactics, techniques, and procedures (TTP) profile, including every command issued, every credential attempted, every lateral path tried, and every protocol used. This is not generic threat intelligence. It is a precise record of how a specific attacker thinks and moves inside your environment.

Security teams use this intelligence to:

- Identify which assets the attacker believed were most valuable, which reveals what your real risks are
- Map the complete movement path and find other potentially compromised accounts or systems
- Improve decoy placement based on how real attackers actually navigate your network
- Inform threat hunting, starting from confirmed attacker behavior instead of hunting blind
- Provide valuable intelligence for planning adversary engagement and red team exercises

Fidelis converts each deception event into a durable record. The intelligence outlasts the individual incident.

- Intelligent Active Deception
- Cyber Resiliency
- Extremely High-Fidelity Alerts
- Proactive Security

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Industries Where Deception Technology Is Critical

Some environments cannot absorb a slow-burn credential breach. Healthcare organizations cannot afford compromised patient systems. Financial institutions face both financial and regulatory consequences from account-level intrusions. OT environments face physical consequences from compromised industrial control systems.

[Fidelis Security](#) has deployed deception for cyber defense for five of the six US military branches and seven of the ten largest US government agencies. The same platform capabilities cover enterprise healthcare, financial services, and critical infrastructure.

In healthcare, deception catches compromised users harvesting patient data over weeks before a single conventional alert fires. In financial services, decoy databases and fake credentials expose attackers who bypassed perimeter controls and are mapping transaction systems. In OT, decoy ICS devices catch reconnaissance that could precede a destructive attack on operational technology.

Is Deception Technology the Missing Layer in Your Security Strategy?

Most security teams already have detection. What they lack is detection they can trust. Fidelis Deception[®] closes the gap. Deception decoys across on-premises, endpoints, network, cloud, and OT. Breadcrumbs seeded directly into Active Directory. Full tactics, techniques, and procedures (TTP) capture on every attacker engagement. Automated response through XDR integration. The result is not more alerts to manage. It is fewer, and each one is real, contextualized, and already moving toward containment before the analyst opens the ticket. Security teams that add deception to their stack stop chasing noise. They start catching attackers, the ones with valid credentials, moving quietly through environments that every other tool called clean.

References:

1. [^Cost of a data breach 2025 | IBM](#)
2. [^2025 Data Breach Investigations Report | Verizon](#)
3. [^IBM Report: Escalating Data Breach Disruption Pushes Costs to New Highs](#)
4. [^The 2024 Study on the State of AI in Cybersecurity | Ponemon-Sullivan Privacy Report](#)
5. [^Industrial Control Systems | Cybersecurity and Infrastructure Security Agency CISA](#)